Abburi Tanusha

ee23btech11201@iith.ac.in | tanusha9461@gmail.com | +91 9032265455 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.37
Tirumala Junior College	Class XII (Board of Intermediate Education)	2023	98.90%
Tirumala EM High School	Class X (Board of Secondary Education)	2021	99.67%

SCHOLASTIC ACHIEVEMENTS AND POSITIONS OF RESPONSIBILITY

- Reliance Undergraduate Scholar
- **UG Buddy(2023-24)** - Sunshine (The Councelling Cell)

Conducted and organized monthly meets with batchmates and seniors of our department.

- **UG Mentor(2024-25)** - Sunshine (The Councelling Cell)

Mentored the freshers assigned and volunteering in many events in the club.

- **Projects Coordinator (2024-25)** - Prakriti (Nature Club of IITH)

PREVIOUS EXPERIENCE

Project Intern, IIT Hyderabad under Prof Dr. Amit Acharyya

- Designed and implemented **Verilog RTL** for 7 FFT architectures (Radix-2 DIT/DIF, Radix-4, Split-Radix, Sliding FFT, MDC Radix-2 pipeline, and 8×8 2D FFT), verified against a Python model with all outputs matching, and analyzed trade-offs across area, timing, and power.
- Synthesized and implemented all designs in Vivado at a 50 MHz constraint with timing met on every design. **Split-Radix** (8-pt) came out best: only **223 LUTs** and **0.117 W on-chip power**, versus 1044 LUTs and 0.134 W for Radix-2 DIT, with scope to reach about 250 MHz.
- Applied pipelining, parallel processing, and folding to 4-tap FIR and 2nd-order IIR filters, analyzing critical path, iteration bound, throughput, latency, and hardware cost; pipelining cut the critical path from 1 multiplier + 3 adders to 1 multiplier + 1 adder, and 4-parallel processing reached 4 samples/clock.

SKILLS

- **Programming Languages** - Verilog, C, Matlab, ARM Assembly, Bash, LaTeX
- **OS** - Windows, Ubuntu
- **PCB Design** - KiCAD, Altium Designer
- **Tools** - LTSpice, Arduino IDE, Cadence Virtuoso, Solid Edge (CAD Simulation), Github, Vivado.

RELEVANT COURSES

- **Core Theory**- Digital Systems, Analog Electronics, Semiconductor Device Fundamentals, Introduction to VLSI Design, Waveguides and Antennas, RTL Design and Verification, Signals and Systems, Control Theory for circuits, Microprocessor and Computer Architecture, Productivity Tools for IC Design, Intro to CMOS Processing, Memory Device Technology for AI/ML, Memory Circuits and System Design, Embedded Systems.

- **Core Lab** - Digital Systems Lab, Semiconductor device fundamentals lab, Circuits lab, Electronic System Design Project Lab, Analog Lab, IC Design Tapeout.

- **Mathematics Courses** - Matrix Theory, Probability, Calculus, Vector Calculus, Differential Equations, Intro to Statistics, Complex Variables

PROJECTS

Wideband Switched-Type Attenuator Design:

- Designed a DC-50 GHz 5-bit switched-type attenuator using hybrid simplified-T, T-type, and -type cells, giving programmable attenuation steps from 0.5 to 8 dB while maintaining impedance matching across the band.
- Derived S-parameter-based equations for attenuation and port matching, and wrote a numerical optimizer in C to solve for the series and shunt resistor values of each cell.
- Simulated and verified all 32 attenuation states in Cadence, achieving a 16.5 dB attenuation range with 3.17° RMS phase error.

Bilinear Interpolation Hardware Design

- Designed and implemented Verilog RTL for the **bilinear interpolation** unit used in the sampling stage of a transformer accelerator for autonomous driving (based on the RETA-AD paper, IEEE TVLSI 2025), exploring multiple architectural approaches to the interpolation datapath.
- Synthesized and implemented the design on FPGA, analyzing area, timing, and power trade-offs between the different implementations.

6-bit Synchronous SAR ADC

- Designed a 6-bit synchronous SAR ADC in Cadence, with a StrongARM latch comparator, a capacitive DAC (CDAC), a bootstrapped sampling switch, and SAR control logic.
- Simulated the full converter and analyzed its performance through metrics such as DNL/INL, SNDR/ENOB, and power consumption.

RTL Synthesis and PPA Analysis

- Developed TCL-based synthesis scripts in **Cadence Genus** for combinational and sequential designs, including adders and a UART, and evaluated power, performance, and area (PPA) across operating frequencies, process corners, and threshold-voltage options.
- **Automated 120 synthesis runs** (10 Verilog designs, 2 frequencies, 6 PDK configurations) with scripted flows, and wrote Python scripts to parse area, power, and timing reports into structured datasets for analysis.

Dynamically Adjusted load Crystal Oscillator:

- Designed a fast start-up Pierce crystal oscillator with an autonomous dynamically adjusted load (DAL) scheme, using a switched capacitor bank to speed up oscillation build-up and improve start-up energy efficiency.
- Simulated the start-up transient in Cadence Virtuoso, with the load capacitance switched after start-up to settle the oscillator at its required steady-state operating point.

Component Tester

- Built an Arduino-based automatic component tester that identifies NPN, PNP, and MOSFET devices, detects their terminal pinout, and estimates the MOSFET threshold voltage by probing the device pins.
- Wrote the measurement and decision logic for automated testing, with extracted device parameters and passive-component values shown in real time on an LCD.

Digital Stethoscope:

- Designed an analog front-end with a TL082 op-amp stage (gain of 150) and on-board RC filtering to amplify and clean low-amplitude heart sounds into a microcontroller-compatible signal range.
- Programmed an ATmega328P for real-time heart-sound acquisition, BPM estimation, and waveform display on a TFT LCD, with a custom KiCad PCB (separate analog and digital sections) housed in a 3D-printed enclosure.

MOSCAP C-V Characteristics at High Frequency:

- Built a numerical device simulator for a MOS capacitor, solving the coupled Poisson and carrier continuity equations under equilibrium, DC bias, and small-signal AC conditions, with Gummel iteration to decouple and converge the nonlinear system.
- Applied Crank-Nicolson time-stepping for the transient AC analysis to extract high-frequency C-V characteristics, and analyzed the accumulation, depletion, and inversion regions.

Digital Clock

- Designed and implemented a Verilog-based digital clock on Vaman Board with 12-hour AM/PM display, using dedicated timekeeping and control logic to drive real-time output.
- Interfaced an LCD module for time display and validated the design on the Vaman FPGA board, confirming correct timekeeping and AM/PM transitions.

EXTRACURRICULAR

- Assisted in executing STEM outreach activities under the Shooting Stars initiative.
- Active NSS volunteer; contributed to walkathons, poster-making campaigns, and social awareness drives.
- Participated in and helped organize various leadership development programs.